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Bluetooth

Communication Protocols

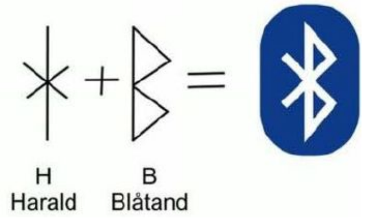
Bluetooth

- A wireless technology used for exchanging data between devices
- A popular short-range wireless communication standard
- Uses UHF radio waves in the ISM radio bands, from 2.400 GHz to 2.4835 GHz
- Used for building personal area networks (PANs)
- Originally developed as a wireless alternative to RS-232 data cables
- Initial development was done at Ericsson Mobile in Lund, in 1989
- The IEEE standardized Bluetooth as IEEE 802.15.1 in 2002
  - But no longer maintains the standard
- The Bluetooth SIG (Special Interest Group) maintains the development of Bluetooth
  - A global community of over 34,000 companies such as Ericsson, IBM, Intel, Nokia, and etc.

Bluetooth

Bluetooth

- Named after the danish king Harald Blåtand (Bluetooth)
  - He was good at communication and was a king that united Denmark
  - The idea behind Bluetooth was to unite different types of communication



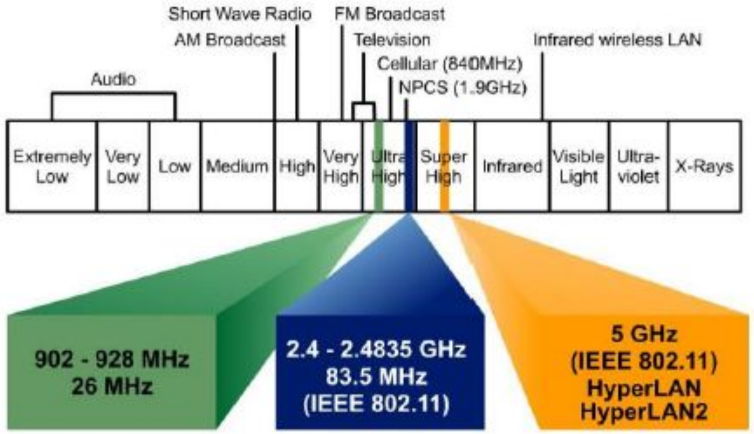
- There are different versions of Bluetooth

Bluetooth 1.x (1998-2003)	**Bluetooth 2.x (2004 - 2007)**	**Bluetooth 3.0 + HS (2009)**	**Bluetooth 4.x (2010 - 2014)**	**Bluetooth 5.x (2016 - …)**
- Base Rate: 1 Mbps - Max Range: 10 m - Latest version: v1.2	- Enhanced Data Rate - Up to 3 Mbps - Max range: 10 m **- S**ecure **S**imple **P**airing - Latest version: 2.1 + EDR	- High Speed mode - Bluetooth v3.0: 3 Mbps - Max range: 10 m - HS Max rate: 24 Mbps - Transmission over WiFi (802.11) connection. - Bluetooth is only used to establish and manage a connection	- Bluetooth v4.2: 1 Mbps - Up to 24 Mbps - Up to 50 m - BLE introduced - Improved privacy to prevent tracking - Almost a new technology - Categories: classic, high-speed, and low-energy - Latest version: v4.2	- 2x speed, 4x range - Up to 48 Mbps - Up to 200 m - Select bands with less interference - Forward Error Correction - Needs new hardware

Classic Bluetooth

Bluetooth

- ❖ Uses ISM (Industrial, Science and Medical) radio bands
- ❖ Frequency ranges: 2400MHz – 2483.5MHz
  - ➢ Power Constrained
  - ➢ License free and free to use
  - ➢ Coexistence: WLAN(802.11), Zigbee(802.15.4), …
  - ➢ Bluetooth divides the bandwidth into 79 channels
    - ■ Each channel has a bandwidth of 1 MHz
    - ■ Guard bands 2 MHz at the bottom end and 3.5 MHz at the top
    - ■ Bluetooth divides data into packets, and transmits each packet on one of 79 channels

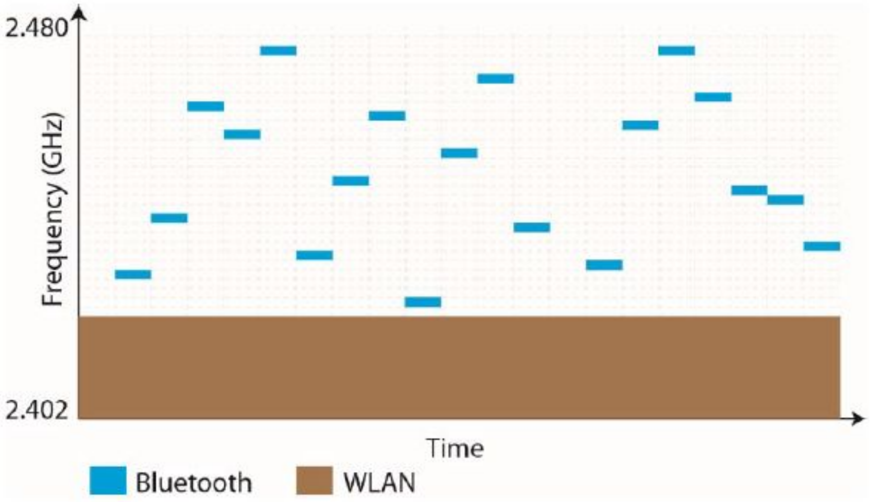


- ❖ Uses Frequency Hopping technology
  - ➢ A technology that spreads the signal over rapidly changing carrier frequencies
  - ➢ Data is sent over a single, changing sub-frequency channel at a time

Classic Bluetooth

- Frequency Hopping / Adaptive Frequency Hopping and Time slots
  - Bluetooth changes the frequency of the data signal 1600 per second
  - Adaptive FH detects interference and dynamically avoids blocked or noisy channels
  - Uses TDMA and each slot length is 625 μs
- The transmit power, and range of a Bluetooth module is defined by its power class
- A module can operate in one or more power classes

	Max. Output Power	Max. Range
Class 1	100 mw	100 m
Class 2	2.5 mw	10 m
Class 3	1 mw	1 m



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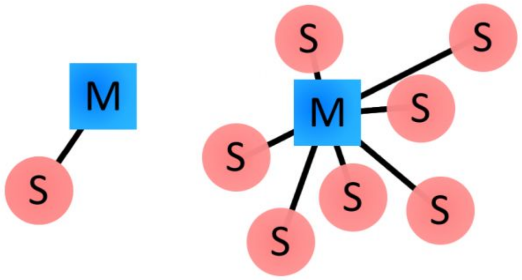
Classic Bluetooth

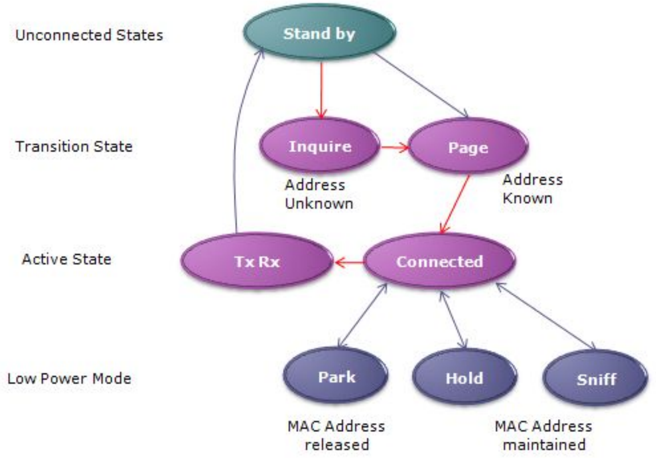
❖ Is a packet-based protocol
   ➢ Every packet is an independent transaction



❖ Master/slave architecture in classic Bluetooth
   ➢ Any slave in the network can only be connected to a single master
   ➢ The slaves can not communicate to each other
❖ A master can communicate with up to seven slaves in a network
   ➢ Slaves get synchronized to the master clock
   ➢ The master's transmission begins in even slots and the slave's in odd slots
❖ Every Bluetooth device has a unique 48-bit address (BD_ADDR)
   ➢ Follows same pattern as LAN MAC addresses
❖ Bluetooth devices can also have user-friendly names (up to 248 bytes long)

Classic Bluetooth

- Connection Process: A multi-step process involving three progressive states
  - **Inquiry** - devices discover each other
  - **Paging** (Connecting) - process of forming a connection
  - **Connected** - After completing the paging process
    - **Active** Mode - The normal mode
      - The device actively transmitting or receiving data
    - **Sniff** Mode - A power-saving mode
      - The device sleeps and only listens to transmissions at a set interval (e.g. every 100ms)
    - **Hold** Mode - A temporary power-saving mode
      - The device sleeps for a defined period and then returns to active mode
      - The master can command a slave device to hold
    - **Park** Mode - The deepest sleep mode
      - A master can command a slave to park
      - The slave will become inactive until the master command it to wake back up



Classic Bluetooth

- Bonding and Pairing
  - Bonds are created through a one-time process called pairing
  - Pairing is a form of information registration for connected devices
    - They share their addresses, names, and profiles, and usually store them in memory
    - They also share a common secret key, which allows them to bond in the future
  - Pairing usually requires an authentication process; e.g. entering a PIN code
  - Bonded devices automatically establish a connection whenever they're close enough
- Bluetooth Profiles
  - Additional protocols, built upon the basic Bluetooth standard
  - More clearly define what kind of data and application a Bluetooth module is transmitting
  - For two Bluetooth devices to be compatible, they must support the same profiles
  - E.g. Serial Port Profile (like RS-232 and UART), Human Interface Device (like keyboards), FTP ...

Classic Bluetooth - Summery

Bluetooth®

- ❖ Reliable - transmissions are based on connected link
- ❖ Relatively high speed, especially with EDR
  - ➢ Suitable for applications which require high data rate and stability
  - ➢ Music / File / Voice
- ❖ High Power Consumption
  - ➢ To perform high speed transmission
  - ➢ To maintain the Link
- ❖ Supports Asynchronous Connection-Less (ACL) transmission
  - ➢ Used for general data packets sent at irregular intervals; supports error retransmission
- ❖ Synchronous Connection-Oriented (SCO) transmission
  - ➢ Used for real-time voice data; uses strictly reserved time slots without retransmissions

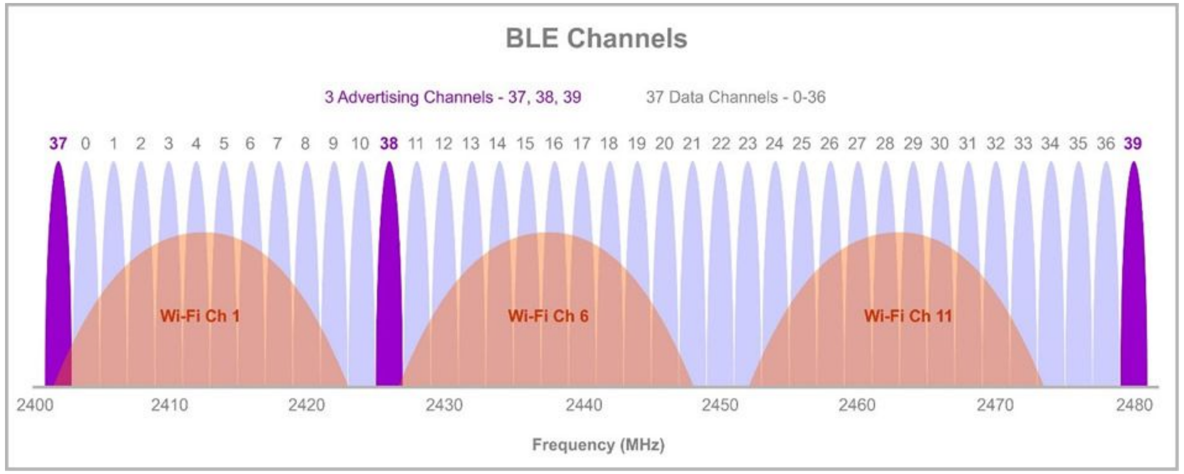
Bluetooth Low Energy - Key Features

Bluetooth®

- A **Wireless Personal Area Network** technology
- **Low-power protocol**: Can run for over a year on a single coin-cell battery.
- **Optimized for short bursts**: Quickly wakes up to transmit data and returns to sleep rather than maintaining a constant active connection.
- **Target applications**: Designed for low-power IoT devices, mobile phones, and tablets.
- **Stateful architecture**: Uses client-server GATT profiles to manage connection states and device attributes.
- **Fast data broadcasting**: Can wake up, advertise its data, and return to sleep within 3 ms.
- **Range & Data Rates**: Maximum range under 100m with data rates of 1 Mbps or 2 Mbps.
- **Hardware advantages**: Small physical size, low cost, and flexible topology options

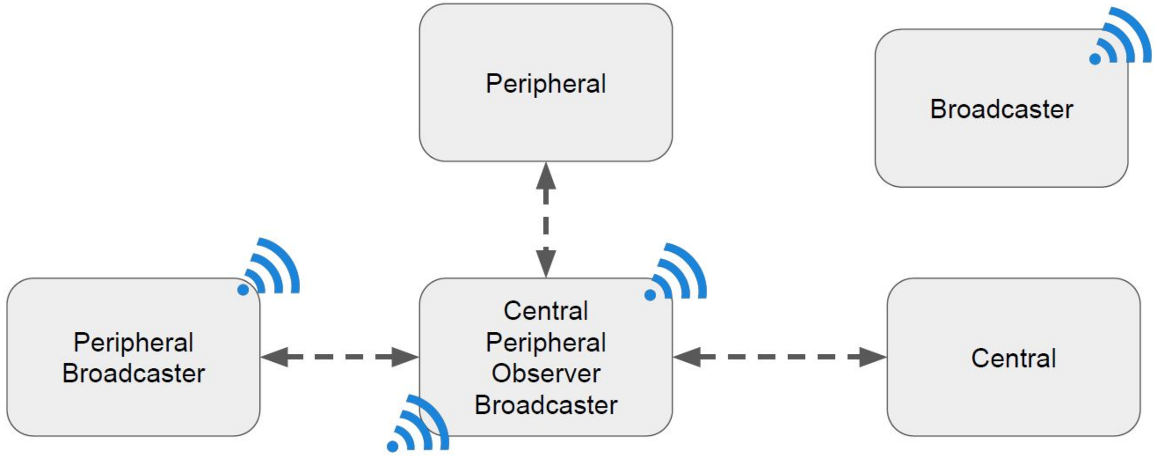
Bluetooth Low Energy - Bands and Channels

- Uses ISM radio bands (2.400–2.4835 GHz)
  - Divided in 40 RF Channels with 2 MHz spacing
  - 3 Advertising Channels
    - Device discovery
    - Connection establishment
    - Broadcasts
    - Selected to minimize interference
  - 37 Data Channels
- Uses Adaptive Frequency Hopping
  - Dynamically identifies interfered channels and removes them from the hop sequence
  - Changes frequencies at the start of every Connection Interval, which ranges from a 7.5 milliseconds up to 4.0 seconds.



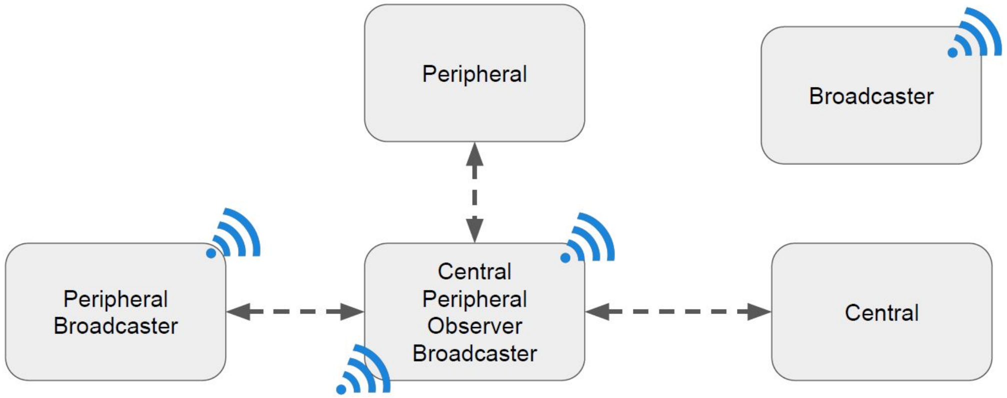
Bluetooth Low Energy - GAP Roles

- All BLE devices use a GAP (Generic Access Profile) to
  - Defines device roles, discovery process, device visibility/management, and connection process between BLE devices.
- The GAP defines four roles:
  - **Broadcaster** (Transmitter only)
    - Can only advertise
      - Sends only advertising packets
    - Can be discovered by observers
    - Cannot be connected
  - **Observer** (Receiver only)
    - Scans for broadcasters and reports the information to an application.
    - Can only send scan requests, but cannot be connected.



Bluetooth Low Energy - GAP Roles

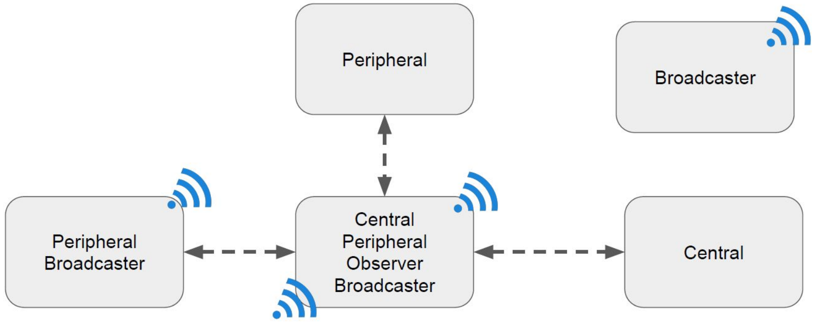
Bluetooth®



- The GAP defines four roles:
  - **Peripheral** - A device that
    - Advertises connectable advertising packets
    - Becomes a slave once it gets connected.
  - **Central** - A device that
    - Initiates connections to peripherals
    - Becomes master when connections are established
    - Supports multiple connections
- A BLE device may support multiple roles
- Each BLE device has a unique 48 bit address which can be
  - **Public:** Follows the standard EUI-48 MAC pattern, obtained from the IEEE.
  - **Random Private**: Rotates dynamically at regular intervals at runtime to prevent device tracking.
  - **Random Static**: Unchanging during runtime, but can generate a new value after a power cycle.

Bluetooth Low Energy - Advertising (Broadcasting)

- **Dedicated Channels:** Three distinct advertising channels (Channels 37, 38, and 39) to broadcast data and advertisement packets.
- **Timing**: Packets are transmitted sequentially across all three channels during an Advertising Interval
- **Configurable Parameters**: Possible to adjust the specific channels used, transmit power, and time intervals



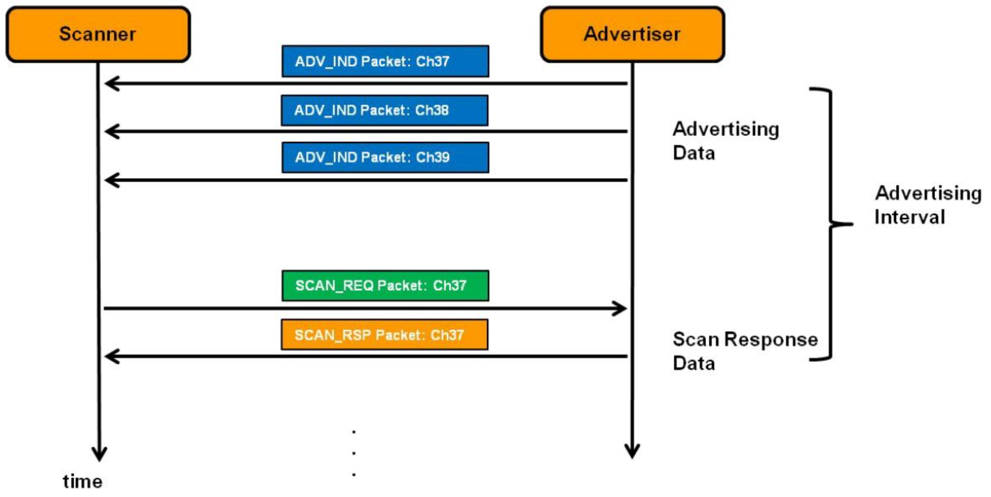
- **Discovery:** BLE devices are detected strictly by listening for these broadcasted packets
- **Scanner Behavior:** A scanning device listens to the advertising channels at regular intervals defined by its Scan Window
- **Scan Request/Response**: A central scanner can request extra data via a "Scan Request" and receive a "Scan Response" back.
  - Exactly one Scan Request/Response pair per event to get extra data without initiating a connection

Bluetooth Low Energy - Scan and Discovery

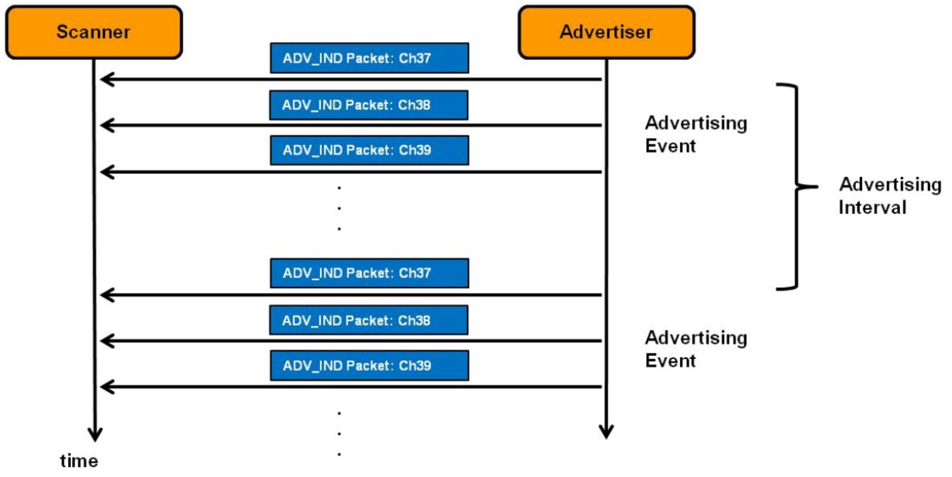
Bluetooth

- Passive Scan
- Active Scan
  - SCAN_REQ: “I want more information”
  - SCAN_RSP: “More information as you wish”

BLE Active scanning

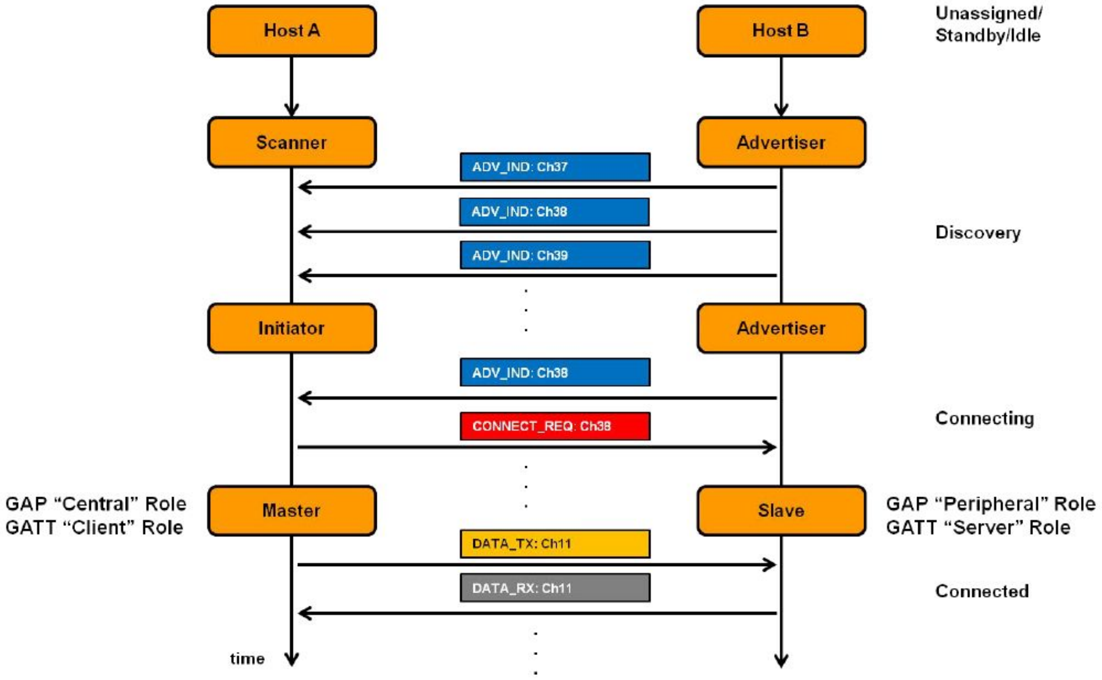


BLE Passive scanning



Bluetooth Low Energy

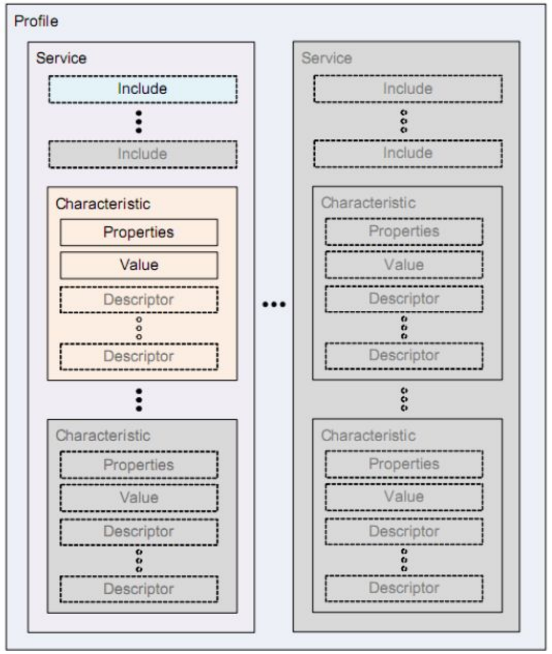
- Scan and Connect
  - CONNECT_REQ: “OK, let’s connect”
    - “Follow these parameters: … … …”
  - Connection Interval (7.5ms – 4s)
  - Connection Timeout (100ms – 32s)



Bluetooth Low Energy - Software Model

Bluetooth

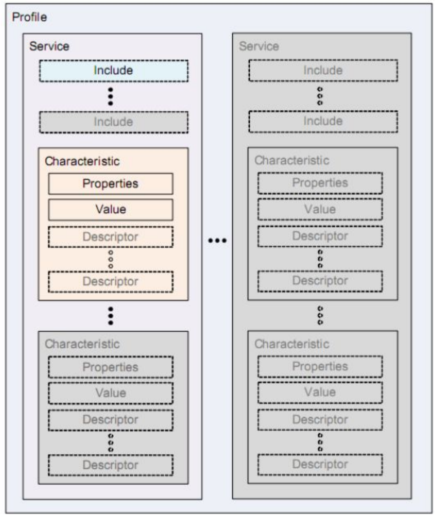
- BLE devices use Generic Attribute Profiles (GATT) to
  - Define how BLE application data is structured and exchanged using services and characteristics, which may be standard or custom-defined
- **Client**: A device that initiates GATT commands and requests, and accepts responses. For example, a computer or smartphone
- **Server**: A device that receives GATT commands and requests, and returns responses. For example, a temperature sensor
- **Characteristic**: A data value transferred between client and server
  - For example, the current temperature.



- **Service**: A collection of related characteristics, operate together to perform a particular function.
  - For example, the ***Health Thermometer*** service includes characteristics for a temperature measurement value, and a time interval between measurements.

Bluetooth Low Energy - Software model

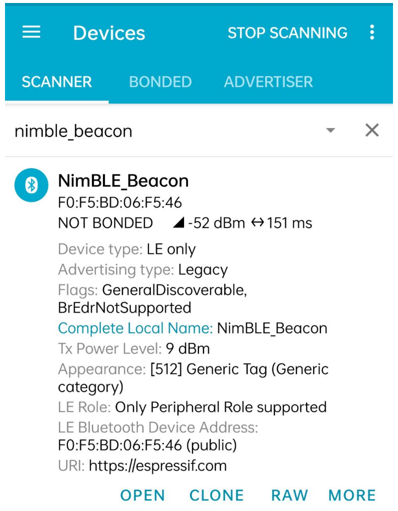
- **Descriptor**: A descriptor provides additional information about a characteristic
  - For example, a temperature value characteristic may have
    - An indication of its units (e.g. Celsius), and
    - The maximum and minimum values which the sensor can measure.
  - Optional – each characteristic can have any number of descriptors.
- Services may also include other services as sub-functions;
  - The main functions of the device are so-called primary services
  - The auxiliary functions they refer to are secondary services
- BLE Identifying applications
  - Services, characteristics, and descriptors are collectively referred to as attributes, and identified by 128 bits UUIDs (Universal Unique Identifier)



Bluetooth Low Energy - Software model

Bluetooth®

- BLE Identifying applications …
  - Broadcasted on advertising channels
  - A random or pseudo random UUID can be used for private use
  - The Bluetooth SIG have reserved a range of UUIDs (of the form xxxxxxxx-0000-1000-8000-00805F9B34FB) for standard attributes
    - For efficiency, these identifiers are represented as 16-bit or 32-bit values
  - Services defined by the Bluetooth SIG



- Data exchange: Read / write of characteristics values; may need authentication
- GATT operations: The GATT protocol provides a number of commands
  - Discover UUIDs for all primary services and find a service with a given UUID
  - Find secondary services for a given primary service
  - Discover all characteristics for a given service and find characteristics matching a given UUID
  - Read all descriptors for a particular characteristic and Read/Write values of characteristics

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Bluetooth Low Energy - Software model

Bluetooth

- GATT operations …
  - GATT offers notifications and indications:
    - The client may request a **notification** for a particular characteristic from the server.
      - The server can then send the value to the client whenever it becomes available.
    - An **indication** is similar to a notification, except that it requires a response from the client, as confirmation that it has received the message
- BLE Security
  - Encryption: AES128 encryption to protect the content
  - Different Types of Key
    - Temporary key: Used in legacy pairing process
    - Short term key: Used to encrypt the legacy key distribution phase
    - Long term key: Replaces short term to encrypt connection
    - Identity resolving key: Used to hide identities
    - Connection signature key: Authentication key

Bluetooth Low Energy

- Introduction
- Device Discovery
- Connection
- Data Exchange

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Bluetooth

- Some useful links
  - [Bluetooth]
  - [Bluetooth Basics]
  - [How Bluetooth Works]
  - [Bluetooth 5: a concrete step forward towards the IoT]
  - [Getting Started with ESP32 Bluetooth Low Energy]
  - [Bluetooth Low Energy App Development: The Basics]
  - [All About Bluetooth - For the Layman]
  - [What is Bluetooth 5? - Gary explains]